

LANDSCAPES OF THE CHURCH

04 — Built Landscapes

Diocesan and Episcopal Conference layers: overview, ArcGIS Online-ready descriptions, methodology, QC, formulas, workflow, and reproducible code

LSAP Integral Ecology Atlas | Technical Documentation | 3 October 2026

9. Analysis 04 — Built Landscapes

9.1 Diocesan layer overview

Layer name	LandscapesChurch_04_Dioceses_Built_V2
Layer alias	04 - Diocesan Built Landscapes
Scale / features	Diocese; 2,874 features
Description	Describes the built-up share of each diocesan geography and pairs it with WorldPop 2025–2030 population change to distinguish urban form from demographic trajectory.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).
Analyses	Built-up share/area; WorldPop 2025/2030; absolute and percent population change; built context; population trend; scale-specific rank/percentile fields.
Key story fields	STORY_BUILT; STORY_CHANGE; STORY_CHARACTER; STORY_CHURCH; STORY_SUMMARY; STORY_COMPREHENSIVE
Status	Publication-ready

9.2 Episcopal Conference layer overview

Layer name	LandscapesChurch_04_Conferences_Built_V2
Layer alias	04 - Episcopal Conference Built Landscapes
Scale / features	Episcopal Conference; 122 features
Description	Combines built-up land and WorldPop population change across Episcopal Conference geographies to describe settlement intensity and demographic direction at a broad Church coordination scale.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).

Analyses	Built share/area; population 2025/2030; change; trend class; Church/LSAP context; population-source interpretation note.
Key story fields	Diocesan story set plus STORY_LSAP and POP_DATA_NOTE.
Status	Publication-ready

9.3 Population fields and formulas

- POP_2025 from TOTAL_WorldPop_2025.
- POP_2030 from TOTAL_WorldPop_2030.
- POP_CHANGE = POP_2030 – POP_2025.
- POP_PCT_CHANGE = $100 \times (\text{POP_2030} - \text{POP_2025}) / \text{POP_2025}$, when POP_2025 > 0.

Population percent change	$\text{POP_PCT_CHANGE} = 100 \times (\text{POP_2030} - \text{POP_2025}) / \text{POP_2025}$ WorldPop-based contextual growth/decline indicator.
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9.4 Built-context thresholds

- <1% — Minimal
- 1–<5% — Light
- 5–<15% — Moderate
- 15–<30% — Highly Built
- ≥30% — Intensely Built

9.5 Population-trajectory thresholds

- <–5% — Strong Decline
- –5% to <–1% — Decline
- –1% to +1% — Stable
- >+1% to +5% — Growth
- >+5% to +10% — Rapid Growth
- >+10% — Very Rapid Growth

WorldPop and ecclesiastical population estimates are displayed as different estimates produced by different sources/methodologies. They are not treated as directly interchangeable denominator counts. POP_DATA_NOTE separates this methodological caveat from the narrative story.

9.6 Workflow

1. Read built-up share and area.
2. Read WorldPop 2025 and 2030 summary fields from the environmental source.
3. Calculate population change and percent change.
4. Classify built context and population trajectory.
5. Calculate scale-specific built and population-growth ranks/percentiles.
6. Write built-landscape and change stories with explicit population-source caveat.
7. Clean schema and retain POP_DATA_NOTE for the Conference derivative.

Relevant reproducible code

A.7 Analysis 04 — Built Landscapes

```
built_pct = safe(row["PCT_LULC_Built"])
```

```

built_area = safe(row["TOTAL_LULC_Built"])

pop_2025 = safe(row["TOTAL_WorldPop_2025"])
pop_2030 = safe(row["TOTAL_WorldPop_2030"])

pop_change = (
    pop_2030 -
    pop_2025
)

pop_pct_change = (
    100.0 * pop_change / pop_2025
    if pop_2025 > 0
    else None
)

def built_class(v):
    if v < 1: return "Minimal"
    if v < 5: return "Light"
    if v < 15: return "Moderate"
    if v < 30: return "Highly Built"
    return "Intensely Built"

def pop_trend(v):
    if v is None: return None
    if v < -5: return "Strong Decline"
    if v < -1: return "Decline"
    if v <= 1: return "Stable"
    if v <= 5: return "Growth"
    if v <= 10: return "Rapid Growth"
    return "Very Rapid Growth"

```

A.10 Safe m² → km² conversion

```

import statistics

values = []

with arcpy.da.SearchCursor(
    feature_class,
    ["ANALYSIS_AREA"]
) as cursor:

    for (value,) in cursor:
        if value is not None and value > 0:
            values.append(float(value))

median_area = statistics.median(values)

if median_area >= 1_000_000:

    with arcpy.da.UpdateCursor(
        feature_class,
        ["ANALYSIS_AREA"]
    ) as cursor:

        for row in cursor:
            if row[0] is not None:
                row[0] = float(row[0]) / 1_000_000.0
                cursor.updateRow(row)

else:
    print("Area already appears to be in km2; no conversion performed.")

```

A.11 Rank / percentile reference implementation

```

def competition_rank(values):
    """
    values = iterable of (feature_id, numeric_value)
    Rank 1 = highest.
    Ties receive the same rank; the next rank skips accordingly.
    """
    ordered = sorted(

```

```

        values,
        key=lambda x: x[1],
        reverse=True
    )

    result = {}
    previous = None
    current_rank = 0

    for position, (feature_id, value) in enumerate(ordered, start=1):
        if previous is None or value != previous:
            current_rank = position

            result[feature_id] = current_rank
            previous = value

    return result

def percentile_at_or_below(all_values, value):
    valid = [v for v in all_values if v is not None]

    if not valid:
        return None

    n = sum(v <= value for v in valid)

    return 100.0 * n / len(valid)

```

A.16 Publication cleanup pattern

```

# Keep only fields intentionally required for the publication derivative.
KEEP_FIELDS = {
    "BUFFER_ID",
    "Account_Name",
    "Country",
    "Language",
    "Official_website",
    "Sector",
    "BUFFER_AREA",
    "LANDSCAPE_SCALE",
    # ... eight PCT_LULC fields ...
    # ... coverage/mosaic/transition fields ...
    # ... story fields ...
}

delete_fields = [
    f.name
    for f in arcpy.ListFields(feature_class)
    if (
        not f.required and
        f.name not in KEEP_FIELDS
    )
]

if delete_fields:
    arcpy.management.DeleteField(
        feature_class,
        delete_fields
    )

# Always run a final required-field check after deletion.

```

Sources and attribution

Ecclesiastical geography and Church statistics: LSAP Integral Ecology Atlas / project Catholic jurisdictional framework; Holy See statistical publications; Catholic-Hierarchy.org. Environmental context: Esri ArcGIS Living Atlas Global Biodiversity and Conservation Hexagons and the underlying credited source organizations.

Esri Global Biodiversity and Conservation Hexagons: <https://www.arcgis.com/home/item.html?id=f1bdbdcac3c54ad8b5b21f5b17d91e8d>

Esri 2026 update: <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/global-biodiversity-and-conservation-hexagons-but-better>

Summarize Within documentation: <https://pro.arcgis.com/en/pro-app/3.6/tool-reference/analysis/summarize-within.htm>